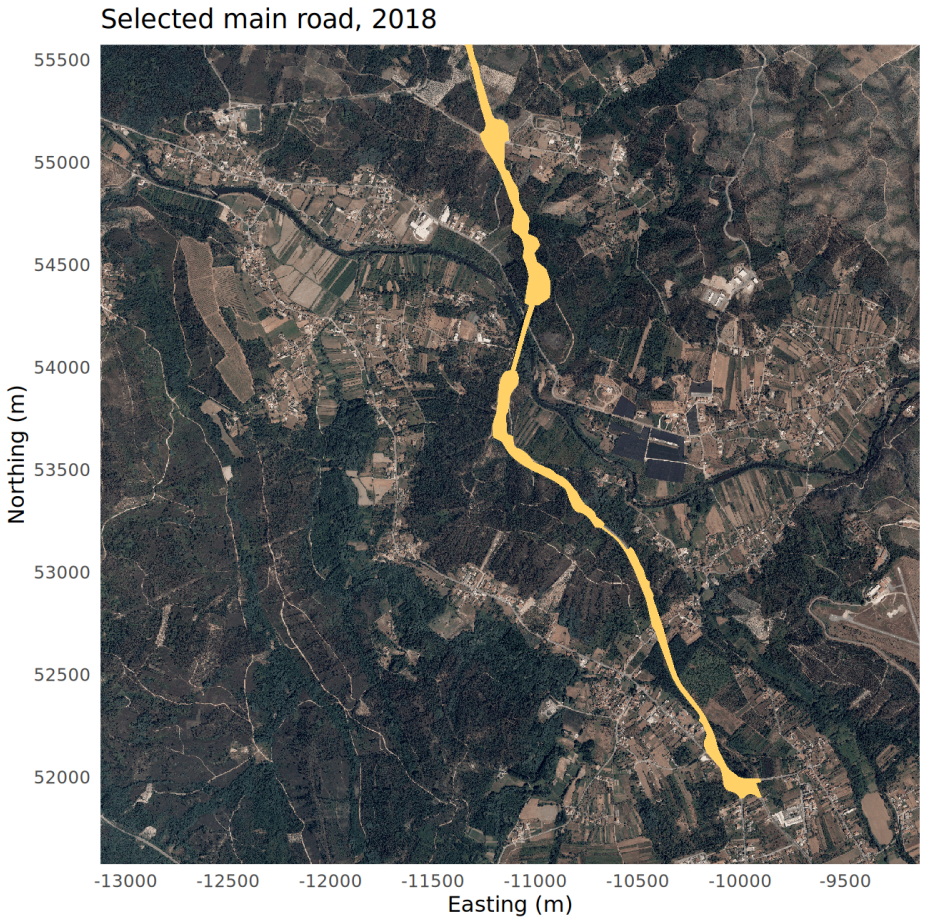
BIOL6130 Final Project: eucalyptus near roads

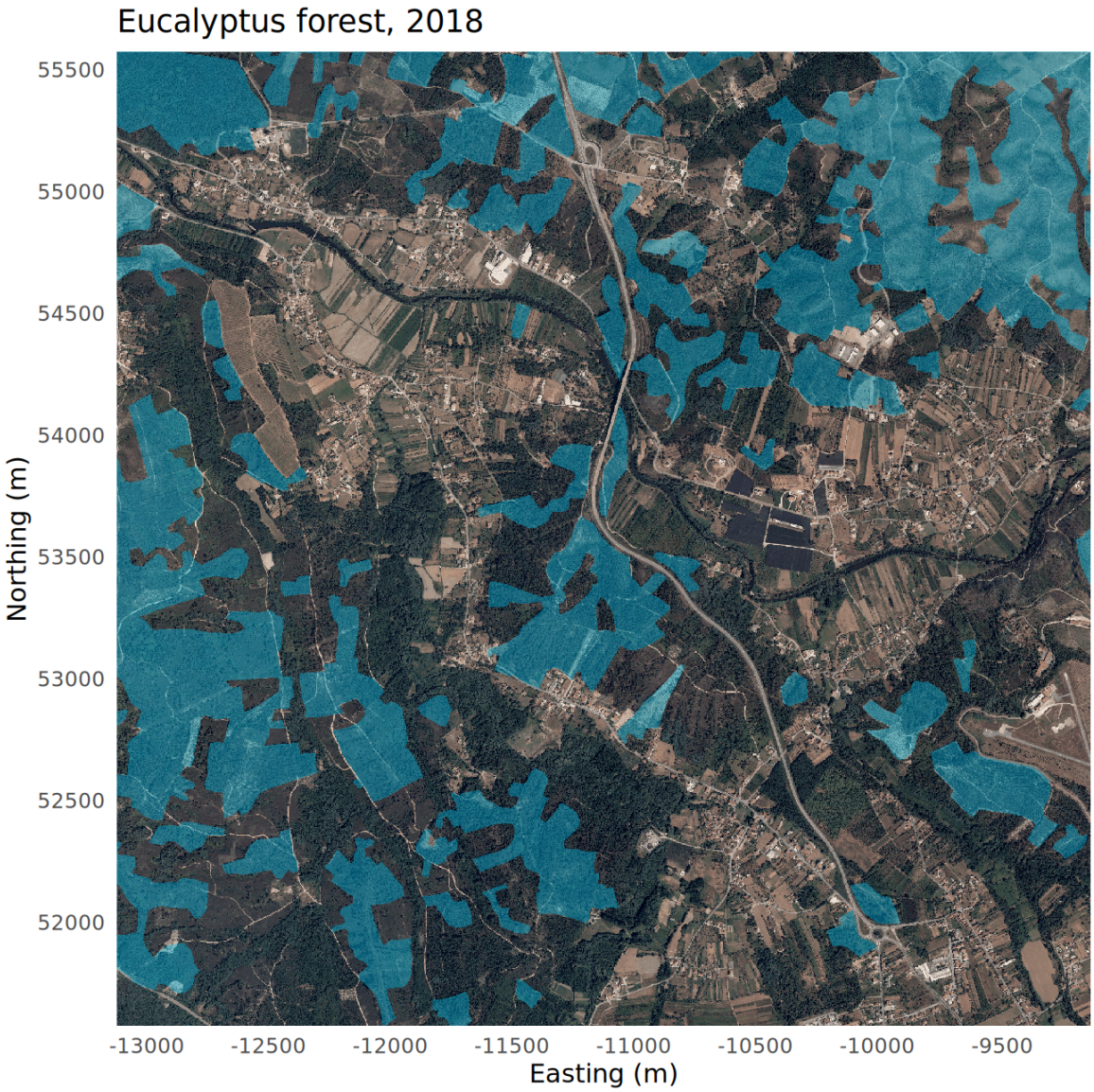
Pilot study in area B

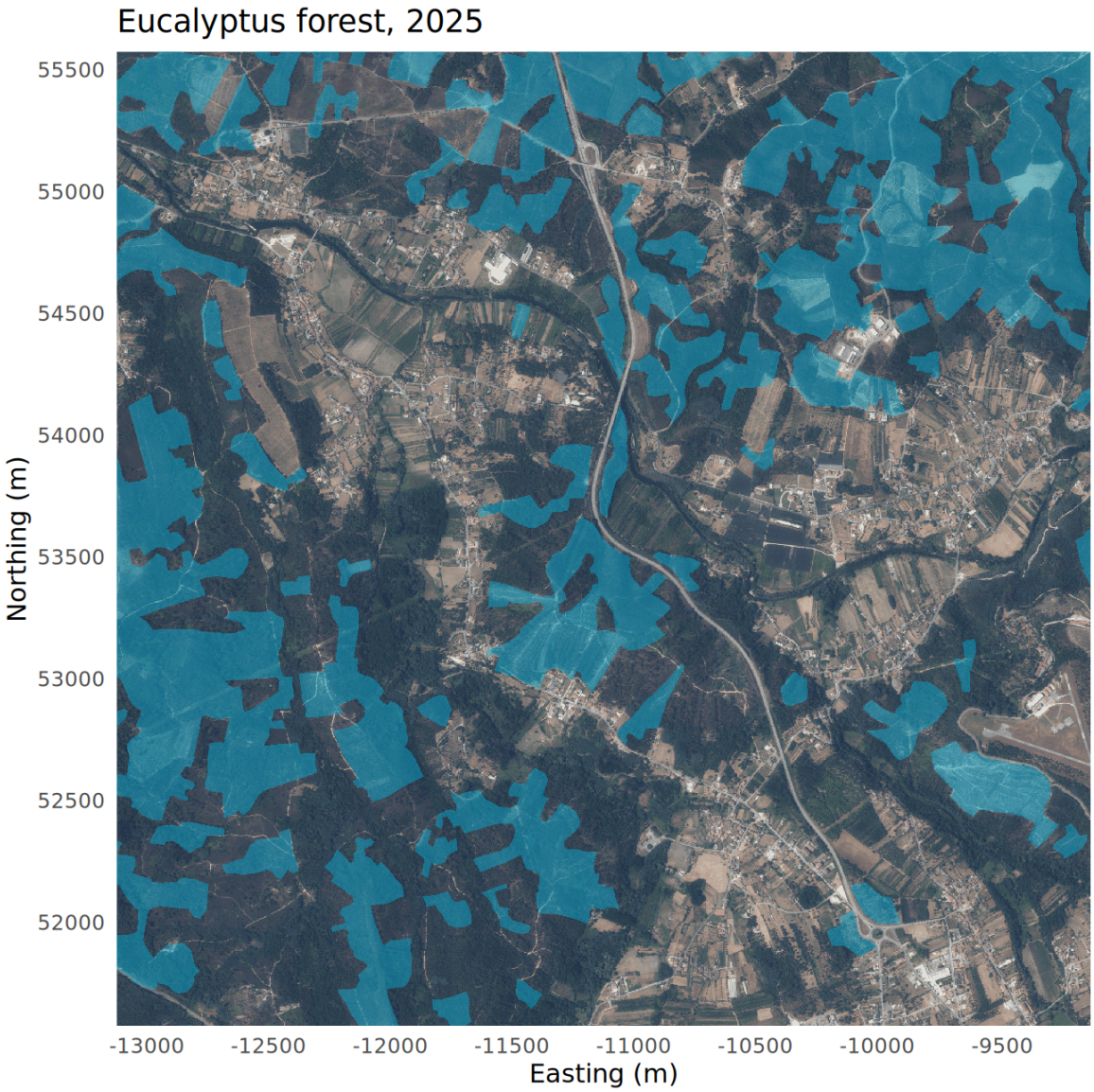
We first looked at a 4 km by 4 km area near Coimbra, Portugal. The yellow line on the first map shows the main road we selected.

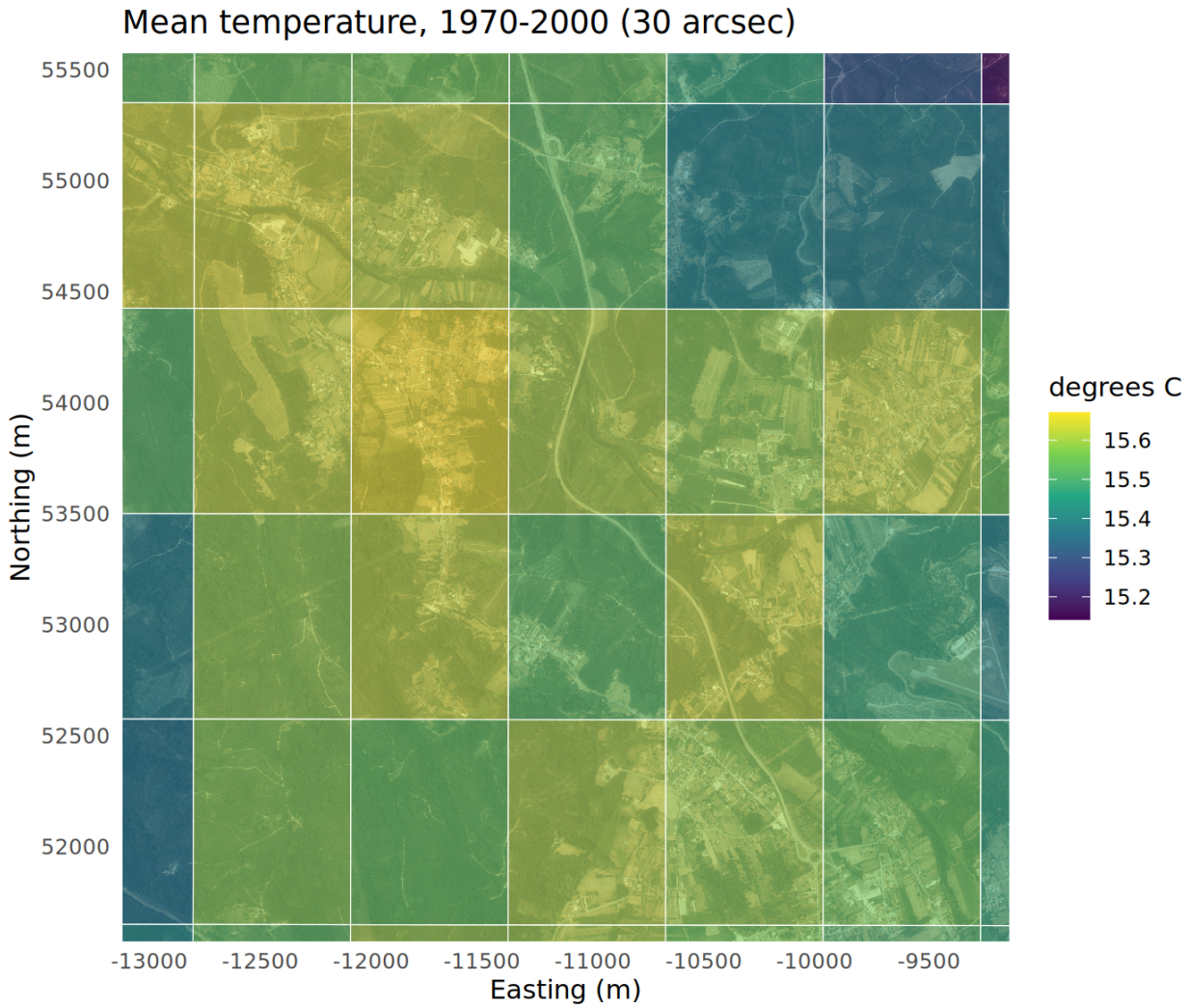
Four fields from the 2018 land-cover map are shown below.

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Rows: 6,063
Columns: 4
$ id         <int> 1288, 1294, 1296, 1297, 1299, 1323, 1361, 1367,~
$ cover_code <chr> "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.~
$ cover_name <chr> "Áreas edificadas residenciais contínuas predom~
$ area_ha    <dbl> 2.5112068, 4.9795026, 1.3266062, 7.0387354, 2.8~
```

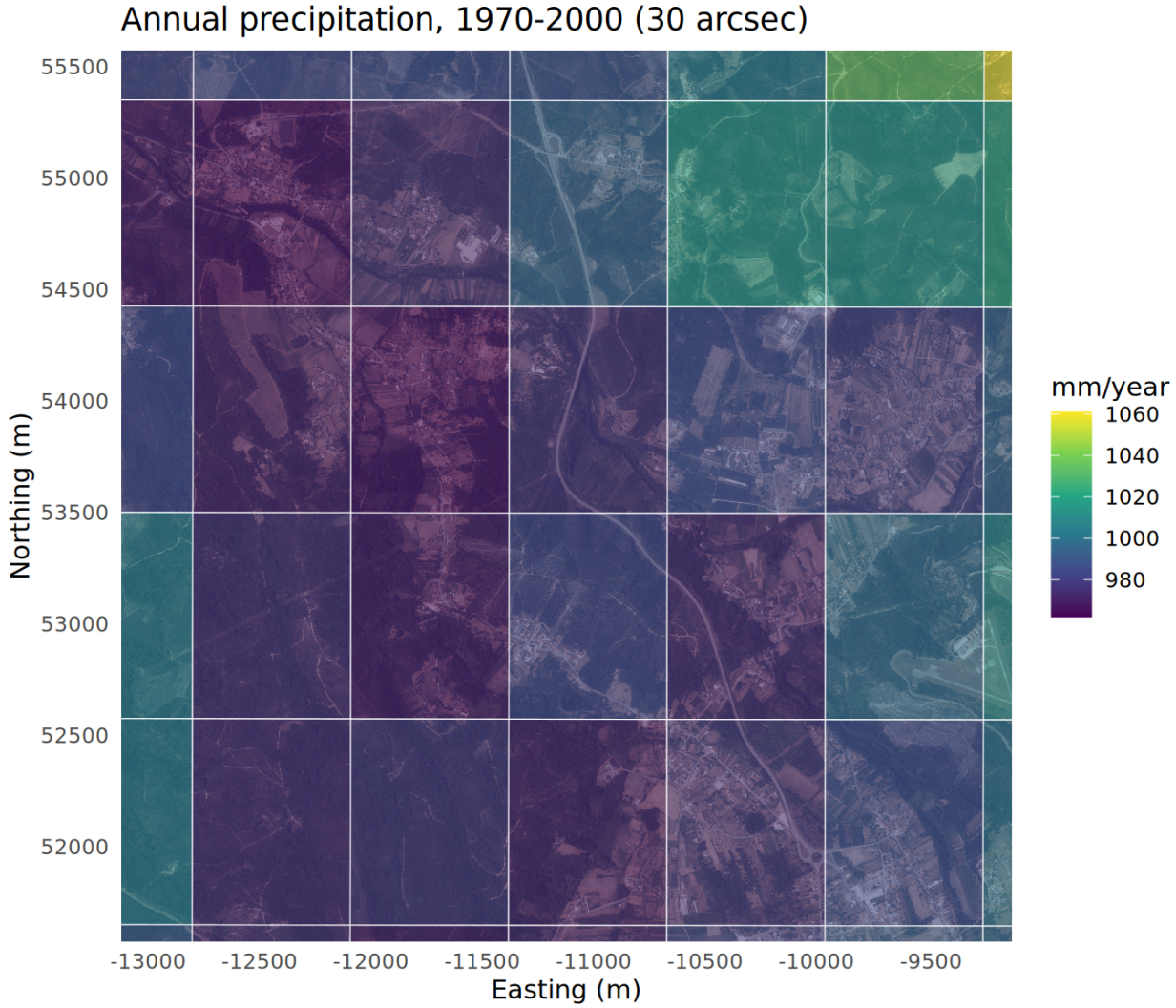



Eucalyptus forest, 2018
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)

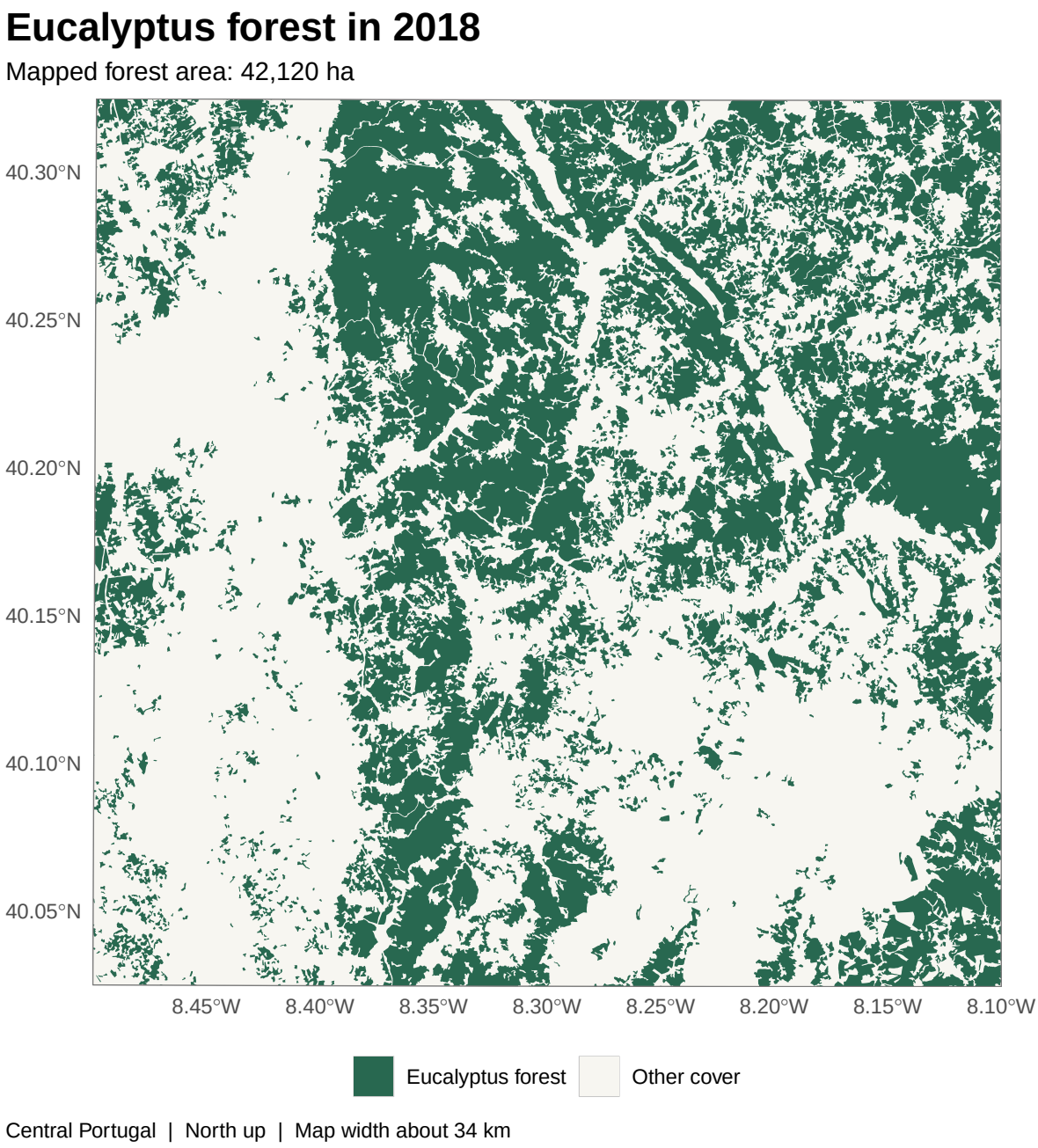

Eucalyptus forest, 2025
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)



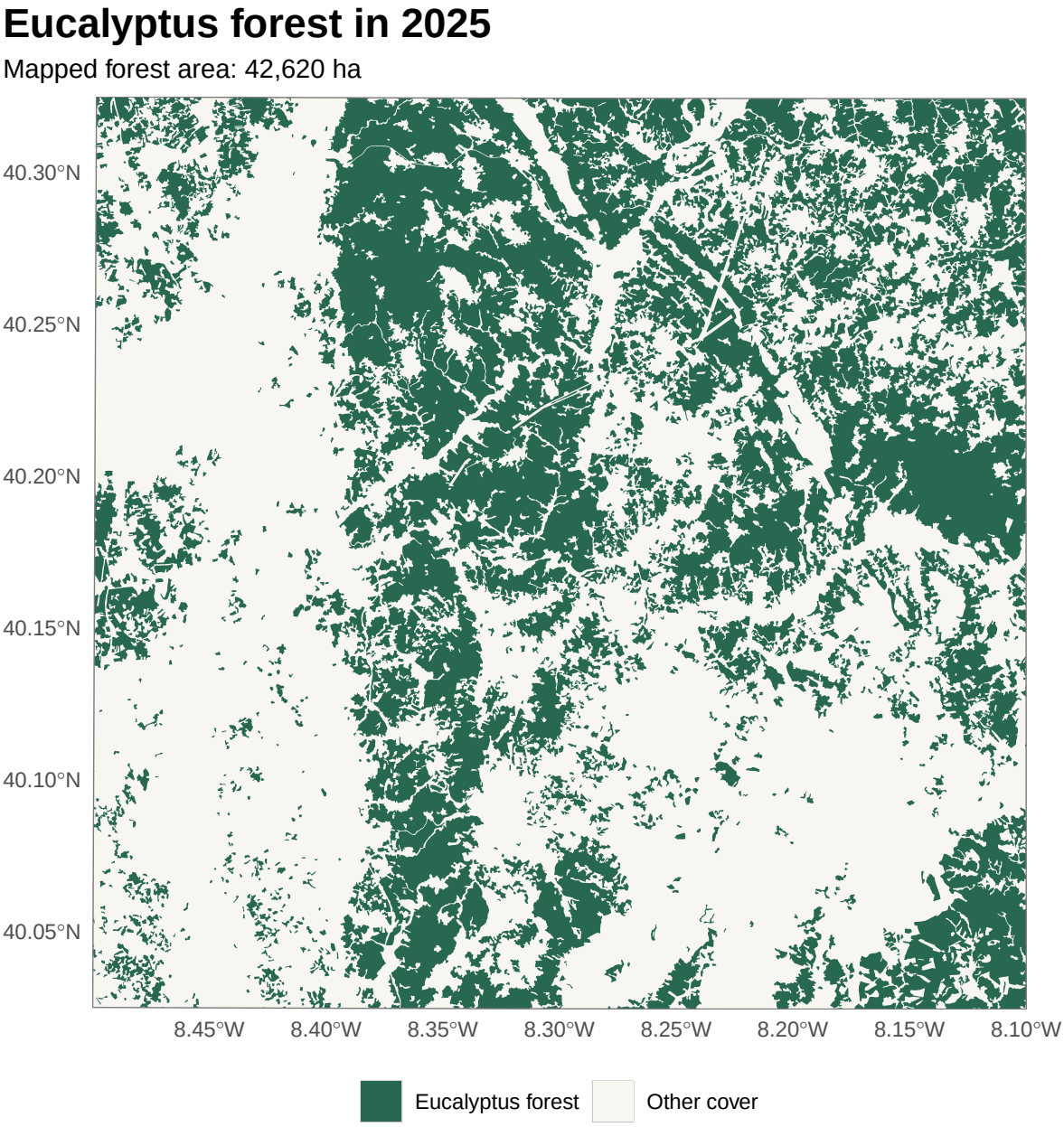
The climate maps show average temperature and annual rainfall for 1970–2000 over the 2025 aerial photo. Each climate cell covers about 710 by 925 m here.


Annual precipitation, 1970-2000 (30 arcsec)
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)
mm/year
1060
1040
1020
1000
980

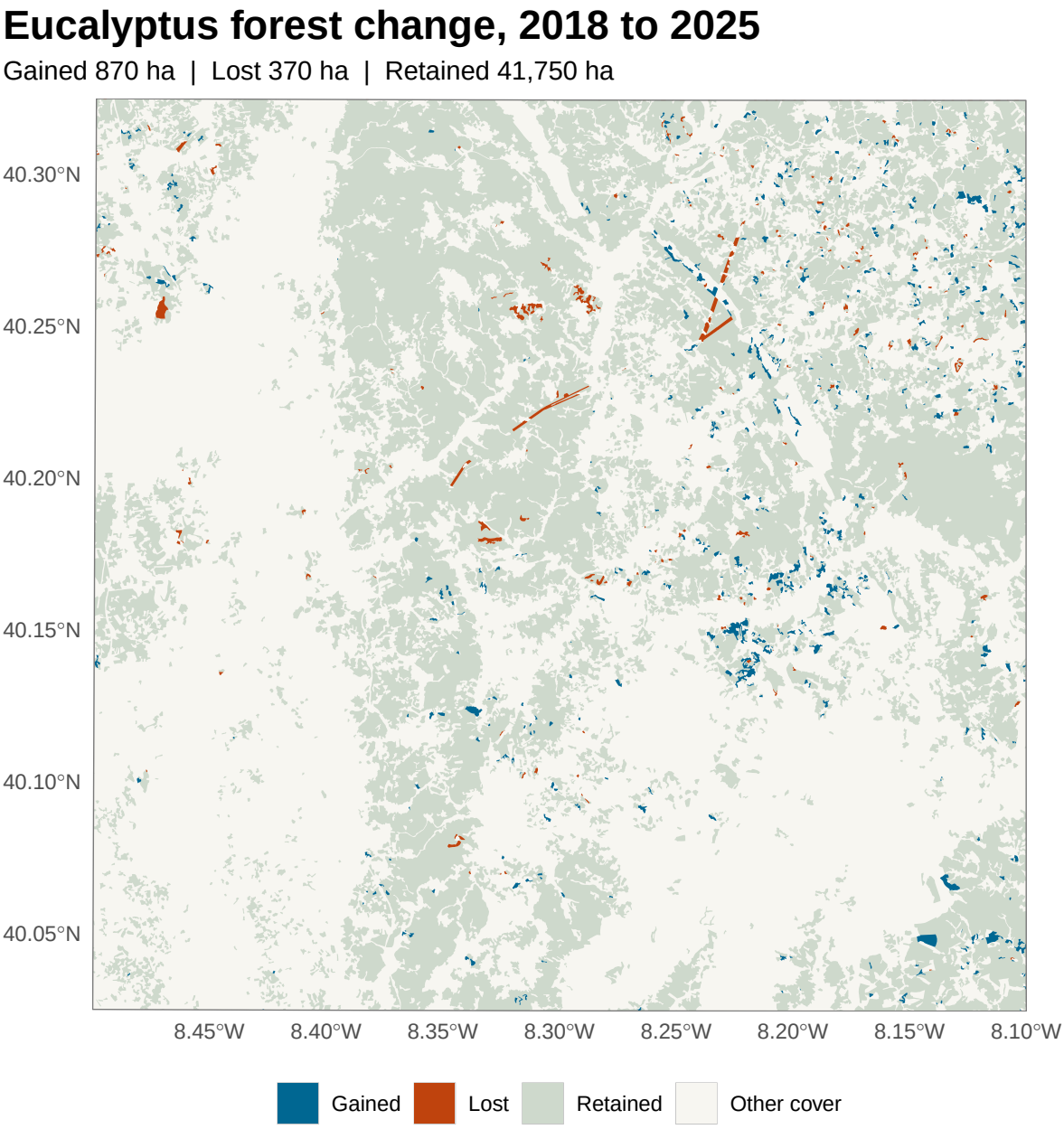
Regional eucalyptus forest cover



Green shows forest mapped as mainly eucalyptus in 2018. The study area is near Coimbra in central Portugal (8.50–8.10°W, 40.025–40.325°N).



The 2025 map shows eucalyptus forest in the same area and at the same scale.



Blue shows eucalyptus forest gained between 2018 and 2025. Orange shows forest lost, and pale green shows forest mapped in both years. The net gain was **500.3 ha**. These maps show the main land cover in each area.

Environment table

Each row in `environment.csv` is one point on the 80 m grid. This was our first environmental overview. The updated comparisons below use actual land areas, organised in 20 m cells. Nearby locations can share the same climate value.

Updated roads

We kept the comparable **2018 and 2025** forest maps and rebuilt the roads from the **January 2018** OpenStreetMap archive. We now include 17,113 mapped line segments in the study area and its distance buffer. Segments are not separate whole roads.

Group	Included routes	Segments
Traffic roads	Motorways, trunk, primary, secondary and tertiary roads and their links; minor connecting, residential and living-street roads	11,308
Land-access routes	4,206 tracks plus 1,599 service segments, including driveways, alleys and service routes without a subtype	5,805

Private and agricultural access restrictions do not remove a physical route, so these routes remain. We exclude parking aisles, drive-through lanes, footpaths, cycleways, steps, pedestrian streets, raceways, unknown road types and unfinished routes. This is a definition for this study; it does not mean the excluded routes have no ecological effect.

A service road is not necessarily a forest road. Of the 1,182 service segments without a subtype, 48% of their length inside the study area crosses artificial surfaces in the 2018 map, and about 26% crosses forests. We therefore call the second group land-access routes. We check results with only tracks, without unknown service subtypes, and without the 29 segments explicitly tagged as closed to motor vehicles.

The two road distances enter the model together. Each contrast changes one road distance while keeping the other observed. The archive can miss routes, and it does not measure traffic or the exact construction date.

Our original predictions. Result 1 asks whether comparable land gains more eucalyptus near roads. Result 2 asks whether gain declines less with distance from old eucalyptus near roads. Both test the broader hypothesis that roads favour forest expansion. Result 3 is an added exploration of pine-to-eucalyptus change in fire footprints.

How we compare land

1. Start with every mapped **2018 eucalyptus source feature**, before looking at 2025 changes.
2. Measure eligible non-eucalyptus land within **100 m of the nearest source**: agriculture, pasture, agroforestry, other forest and shrub cover. Exclude artificial surfaces, bare land, wetlands and water. Use 200 m as a wider check. Count each piece of land once.
3. Divide this land into 20 m accounting cells, retaining the **actual area** of each original land-cover fragment. Use a point inside the fragment for distances and environmental values. The cell is not one tree, and it does not improve the accuracy of the original maps.
4. Model the fraction of each piece mapped as eucalyptus in 2025, weighted by its eligible area. Compare variation within the **same source feature and original land-cover class**. Also include source distance, both road distances, elevation, slope, temperature, rain, wind, measured 2017 fire history and the 2015 forest-management label.

The main model includes 1727 source features, 26308.5 ha of eligible land and 766.0 ha of mapped gain. Source features can touch or contain several parts; they are not necessarily separate biological stands.

A fair comparison needs available land on both sides. For each road type we compare the area-weighted lower and upper quarters of baseline road distance:

Road	Nearer (m)	Farther (m)
Traffic roads	72	361
Land-access routes	69	387

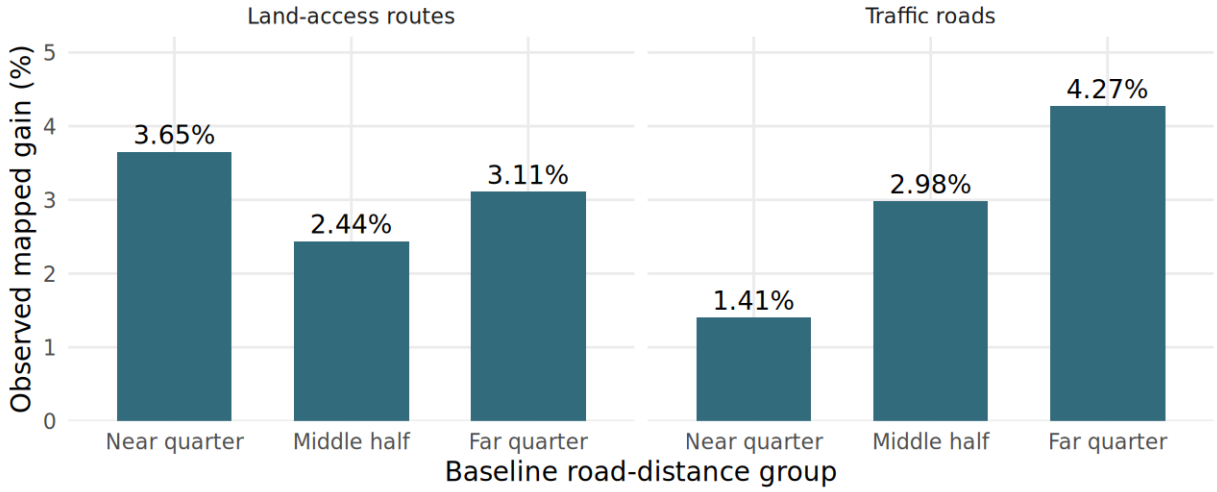
For the reported contrasts, a source/cover group must span both distances **within a similar distance range of the other road type**. Those other-road ranges are four baseline distance bands. Result 2 additionally requires locations from 25 to 75 m outside the old forest at both road-distance ends. We report the resulting reference areas with each result. These checks reduce extrapolation; they do not make every environmental condition identical.

The model uses log(1 + distance), with each road distance interacting with source distance. Other continuous environmental adjustments are linear. The main linear model includes all eligible land, including groups with no gain. It estimates adjusted differences, not absolute conversion probabilities. A bounded fractional-logit model checks this choice on groups with some conversion.

The 100 m window follows the scale used in an earlier eucalyptus mapping study. The 200 m check, road quartiles and 25-to-75 m comparison are analysis choices, not known limits to seed travel. Climate values are long-term averages, not local weather during dispersal.

Result 1

Is mapped eucalyptus gain greater near roads? First, the observed percentage uses new eucalyptus area divided by all eligible land in that distance band.



These bars do not control for the surrounding land. After adjusting for the source forest, original cover, measured environment and the other road type, the **nearer-minus-farther** differences were:

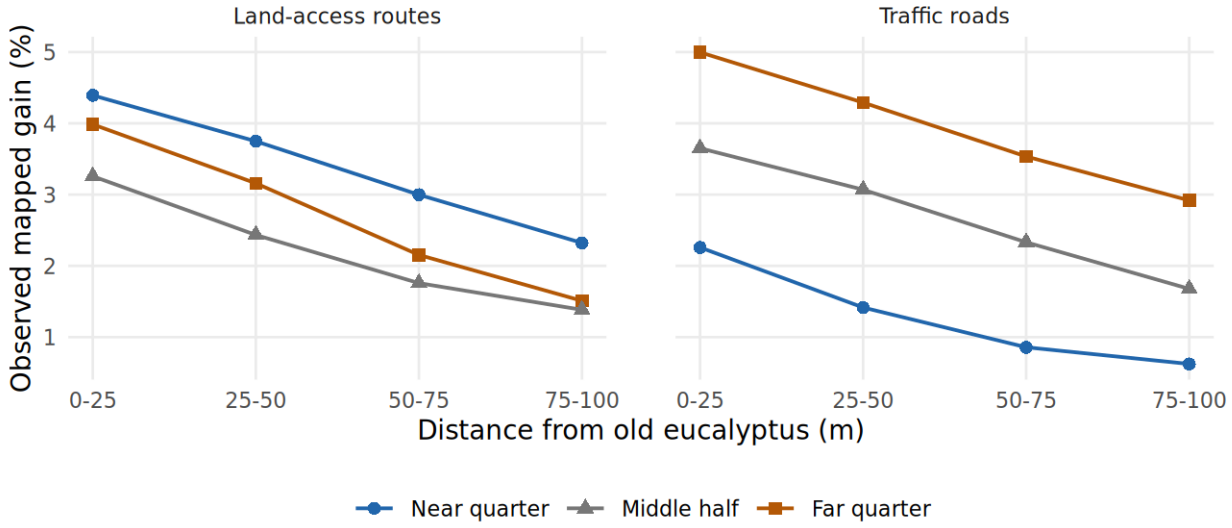
Road	Difference (pp)	95% interval	Sources	Area (ha)
Traffic roads	-0.17	-0.72 to 0.38	222	11696
Land-access routes	-0.18	-1.03 to 0.68	153	10576

Positive values would support more gain nearer roads. Both estimates are small and negative, but **both uncertainty intervals include zero and positive values**. The data therefore do not give clear support for greater gain near either road group. They also do not establish that roads reduce gain or have no effect.

The two rows use different supported reference areas and reference distances. Their effect sizes should not be ranked as if they were the same experiment. The area denominators account for the fact that more land may be available far from a road.

Result 2

Does gain decline less with distance from old eucalyptus near roads? The graph below shows observed change before environmental adjustment.



For the adjusted test, we measure the change from **25 to 75 m outside old eucalyptus** at each reference road distance. We then subtract the farther-road change from the nearer-road change. A positive value agrees with the prediction that mapped gain declines less near roads.

Road	Difference (pp)	95% interval	Sources	Area (ha)
Traffic roads	+0.06	-0.12 to 0.24	79	6543
Land-access routes	-0.07	-0.23 to 0.10	48	7268

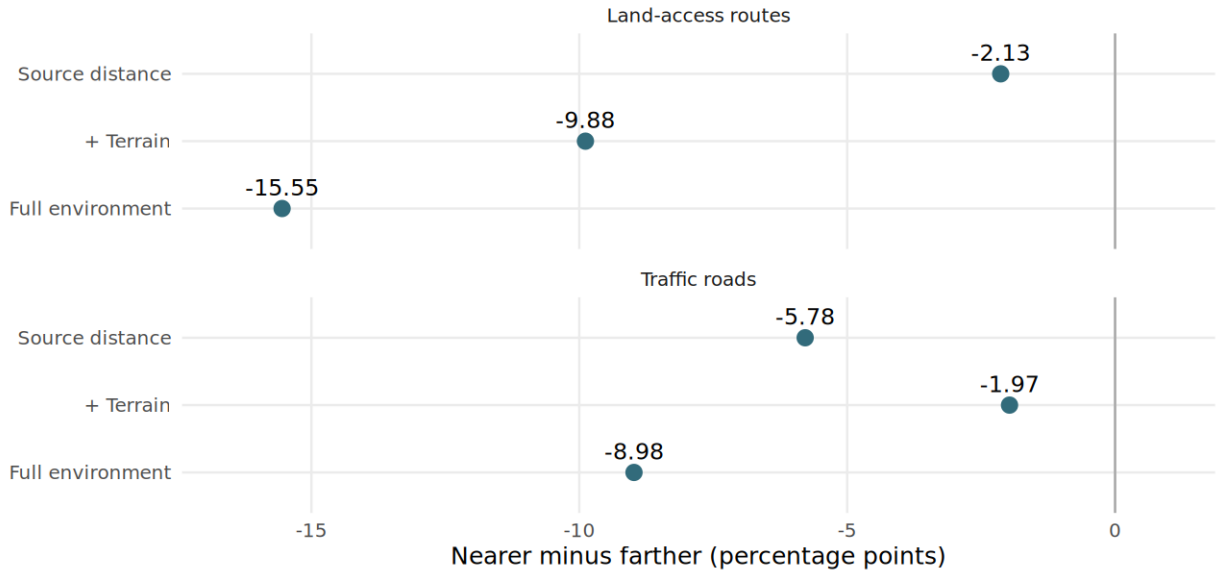
The traffic-road estimate is slightly positive and the access-route estimate slightly negative. Both intervals cover zero and both directions. **Neither gives a clear road-related difference in the distance pattern.** This describes mapped forest conversion, not how far individual seeds travelled.

Result 3

On baseline pine land covered by fire, was eucalyptus conversion greater near roads?

The same 18 fire footprints from 2019–2024 overlap **176.23 ha of maritime-pine land** mapped in 2018. In 2025, **17.33 ha** is mapped as eucalyptus. Only **three fires** contain this change. The model compares places within each of these fires, controlling both road distances and distance to old eucalyptus. The other 15 fires remain in the area totals but cannot estimate a finite within-fire logistic relationship.

For traffic roads the comparison is 132 versus 470 m; for land-access routes it is 112 versus 595 m. These quartiles come from all baseline burned-pine land. Each contrast keeps the other road distance observed and uses fires spanning both distances. Each displayed comparison is supported by only two fires.



These estimates do not show more conversion near either road group. However, **one 2019 fire contains 85.8% of the change**. Removing it reverses the traffic-road difference to +17.99 percentage points; only one remaining fire supports that distance contrast. We report no cell-based confidence intervals or p values: three changing fires are too few for reliable regional inference.

The terrain check adds elevation and slope. The full check also adds climate, the 2015 planting label and 2017 fire severity, using 148.21 ha with complete data. Current-fire severity is missing. These maps cannot establish whether planting, mixed stands or natural recruitment explain the change, or whether the pine-to-eucalyptus switch happened before or after the fire. There is no unburned control for a fire effect.

Checks and limits

The following checks keep the same baseline reference distances. All entries are percentage-point differences. T = traffic roads; A = land-access routes. Result 1 is nearer minus farther. Result 2 is the difference in the 25-to-75 m source-distance change.

Analysis	T1	A1	T2	A2
Main, 100 m	-0.17	-0.18	+0.06	-0.07
Wider, 200 m	-0.23	-0.28	+0.08	-0.22
Tracks only for access	-0.17	-0.15	+0.06	-0.07
No unknown services	-0.17	-0.15	+0.06	-0.07
No motor bans	-0.17	-0.18	+0.06	-0.07
Bounded, changing groups	-0.65	-0.68	-0.38	-0.09
Linear, same groups	-0.50	-0.62	-0.27	-0.26

Changing the road definition did not reveal a stable positive relationship. The bounded model uses only 367 source/cover groups with some conversion: 7,575 ha from 291 source features. The last row fits the linear model to exactly that same land. This separates a change in sample from a change in model. The bounded row gives point estimates only.

The main linear model puts **40.4%** of fitted land area outside the 0–1 range. This limits its use to adjusted differences, not absolute predictions. We did not clip predictions. The distance relationships and environmental adjustments are model assumptions, not experimentally verified controls.

Uncertainty allows shared source features and shared 2 km geographic blocks. Using 1 km and 4 km blocks also left all four primary intervals spanning zero. The 4 km covariance calculation needed a numerical correction, which widened the traffic-road intervals. The main 2 km calculation did not need it. These block sizes handle correlated errors; they are not seed-dispersal distances. Deleting one source at a time gave Result 1 ranges of -0.26 to +0.05 points (traffic) and -0.57 to -0.06 (access). Result 2 stayed +0.04 to +0.11 (traffic) and -0.10 to -0.02 (access). These are influence checks, not uncertainty intervals.

Several explanations remain open. Road access may support planting, harvesting and replanting as well as natural recruitment. Pine-labelled forest can already contain eucalyptus; fire and resprouting may change the dominant cover. Farming and abandonment, soils, drainage, weather during seed release, road maintenance and traffic are not fully measured. The coarse 2015 planting label does not verify whether a particular new patch was planted.

Seven years can reveal mapped cover change, but not every new seedling becomes detectable. COS includes young stands capable of reaching forest conditions and does not count individual trees. Across the road definitions, distance windows and models checked here, we did not find stable support for either original prediction. Some estimates point in the predicted direction, but uncertainty or model sensitivity prevents a clear conclusion. This does not cover every possible method or show that roads have no effect. The maps measure forest-cover change, not natural seed dispersal itself.

Data sources

- **Land cover and the pilot road.** Direção-Geral do Território (DGT), COS Series 2, 2018 v4 and 2025 v1. The maps use eucalyptus class 5.1.1.6 and maritime pine class 5.1.2.1. The regional `pine2025.gpkg` file contains the 2025 maritime-pine polygons clipped to the same study area. DGT class definitions and imagery dates. DGT, CC BY 4.0
- **Regional roads.** The original network is retained in `roads.gpkg`. The revised `roads2018_selected.gpkg` comes from the 1 January 2018 archive and separates traffic roads from tracks and selected service roads. Private access is retained. © OpenStreetMap contributors. Geofabrik archive; track, service and access definitions; ODbL 1.0.
- **Pilot aerial photos.** DGT imagery from 2018 and 2025, exported at 2 m for area B. DGT orthophotos, CC BY 4.0
- **Pilot climate.** WorldClim v2.1, temperature and rainfall averages for 1970–2000. WorldClim · Fick and Hijmans (2017)
- **Regional climate.** CHELSA v2.1, temperature, rainfall and mean near-surface wind speed for 1981–2010. CHELSA · BIOCLIM+ data, CC0
- **Elevation and slope.** Copernicus DEM GLO-30, tile N40/W009. Product and licence
- **Fire.** ICNF, *Áreas ardidas*, 2017–2025, retrieved 5 November 2026. Fire perimeters · ICNF terms. The 2017 burn severity and image quality layers come from Jahanianfard et al. Portuguese Burn Severity Atlas v2, CC BY 4.0
- **Forest management.** Lesiv et al., global forest management map for 2015, v3.2. Data, CC BY 4.0 · Map validation and limits
- **Planting records.** ICNF public RJAAR projects and national summary of approved actions.

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Research papers

- **Field distance.** Deus et al. (2019). Wild eucalyptus at and beyond plantation edges in Portugal. Biological Invasions
- **Source distance in a mapping study.** Deus et al. (2022). Potential eucalyptus recruitment in Iberian protected areas. Forests
- **Fire and pine.** Águas et al. (2014). Natural establishment of eucalyptus in burned stands in Portugal. Forest Ecology and Management
- **Fire and resprouting.** Catry et al. (2013). Survival and regrowth of planted blue gums after fire in Portugal. Forest Ecology and Management
- **Pine recovery.** Maia et al. (2012). Fire severity and maritime-pine regeneration in central Portugal. Annals of Forest Science
- **Seedling habitat.** Fernandes et al. (2018). Habitat disturbance and wild trees around Portuguese plantations. Biological Invasions
- **Farm and forest history.** Nunes et al. (2023). Land-use change in Lousã, central Portugal. Quaestiones Geographicae
- **Statistical methods.** Papke and Wooldridge (1996); fixest 0.14.2 documentation.